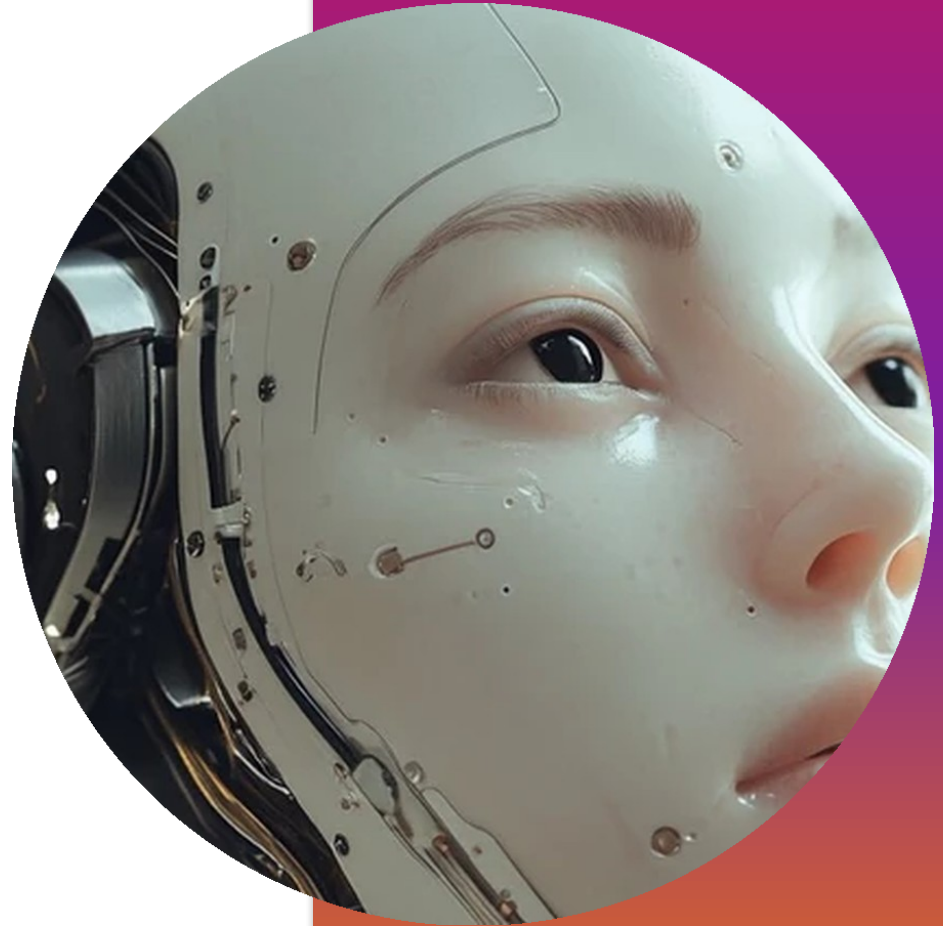


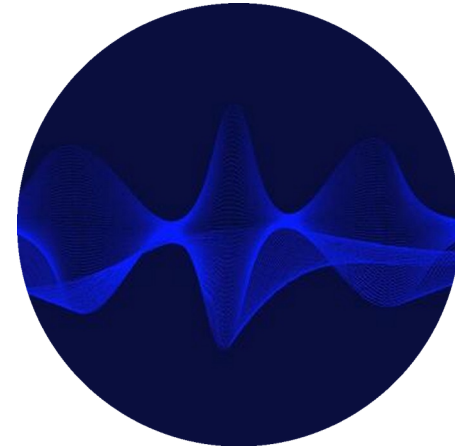
AI CONFESSOR

BY:

SAI KOUSHIK VIMMADisetty



AGENDA



- Introduction
- Proposed method
- Techniques used
- Results
- Summary
- References

INTRODUCTION

- Real-time conversational AI: a system you talk to, and it talks back
- The problem: text-only chatbots feel slow, rigid, and unnatural
- Voice is the most natural interface — no typing, no waiting
- Goal: an always-on voice companion that listens, thinks, and speaks



WHAT IT IS

- AI Confessor: a voice-first conversational AI web app
- FastAPI backend (Python 3.12) + React + TypeScript + Vite frontend
- Docker Compose: backend + frontend behind nginx
- Live on AWS with Anthropic Claude (claude-sonnet-4-6) as the brain

PROPOSED METHOD: ARCHITECTURE

Mic → WebSocket → FastAPI → Vosk → LLM → Edge TTS → Speaker

- Audio is captured in the browser and sent over a WebSocket
- FastAPI transcribes the speech locally with Vosk
- The text goes to the LLM; tokens stream back to the user live
- Edge TTS converts the reply to speech for the speaker



WEBSOCKET PROTOCOL

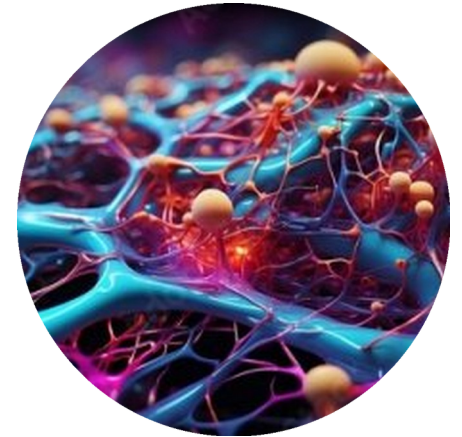
- One persistent WebSocket connection per conversation session
- Session handshake on connect; ping/pong keeps the socket alive
- Server pushes token deltas as the LLM generates them
- Audio chunks in, text plus audio out — on the same socket

SESSION ISOLATION

- Every session keeps its own conversation history
- Sliding window trims old turns — the system prompt is always preserved
- Concurrent users never see each other's history
- Isolation verified with multiple simultaneous sessions

TECHNIQUES USED: LLM PROVIDERS

- Pluggable providers, selected by the LLM_PROVIDER setting
- anthropic — live deployment runs claude-sonnet-4-6
- openai — supported with an OpenAI key
- mock — offline provider for tests and QA



TECHNIQUES USED: VOSK + EDGE TTS

- Vosk: speech recognition that runs fully locally — no cloud, no key
- Model `vosk-model-small-en-us-0.15` auto-downloads on first run
- Edge TTS synthesizes voice replies server-side as mp3
- Audio streams back over the WebSocket; the browser plays it through an audio queue
- Voice toggle in the UI controls spoken replies



TECHNIQUES USED: AWS DEPLOYMENT

- EC2 t3.medium in us-east-1 running Ubuntu 24.04
- App runs in Docker Compose: backend + frontend behind nginx
- Free public HTTPS via a Cloudflare quick tunnel pointed at the frontend (HTTPS is required for phone microphone access)
- The Claude API key lives server-side only in the backend .env (chmod 600) — never in frontend code or Git
- Deployment automated with EC2 user-data, which publishes the tunnel URL through the frontend container
- Code pushed to GitHub (AI-Confessor repo)



RESULTS: QUALITY ASSURANCE

- 12/12 mock WebSocket checks passed
- 8/8 pytest suite passed
- Concurrent-user sessions stay isolated
- Vosk transcribed a 3.4 s clip exactly: “the quick brown fox jumps over the lazy dog”
- Frontend production build: clean
- Secrets scan clean — no keys in code



RESULTS: LIVE AWS TESTS

- Page loads over HTTPS with no interstitial
- Header shows claude-sonnet-4-6 connected (not mock)
- Live chat test: the AI replied “PONG” exactly to a one-word instruction — proving live Claude, not canned replies
- Microphone capture and speech-to-text confirmed working from an iPhone
- Live URL: <https://passed-tampa-single-efforts.trycloudflare.com> (quick-tunnel URLs change on redeploy)

HONEST LIMITATIONS

- The Cloudflare quick-tunnel URL is ephemeral — it changes on every redeploy (fine for a demo; a custom domain plus a named tunnel for production)
- Edge TTS is a network service, so voice replies depend on connectivity
- Single EC2 instance — no high availability yet

SUMMARY

- Built: a voice-first conversational AI — FastAPI + WebSockets + React
- Deployed: live on AWS (EC2 + Docker + Cloudflare HTTPS tunnel) with Claude claude-sonnet-4-6
- Verified: QA suite green, live PONG test proves real Claude, iPhone mic/STT works
- Secure by design: API keys live server-side only, never in frontend code or Git



REFERENCES

- FastAPI — <https://fastapi.tiangolo.com/>
- Vosk — <https://alphacephei.com/vosk/>
- edge-tts — <https://github.com/rany2/edge-tts>
- Anthropic Claude — <https://docs.anthropic.com/>
- React — <https://react.dev/>
- Docker — <https://www.docker.com/>
- AWS EC2 — <https://aws.amazon.com/ec2/>
- Cloudflare Tunnel — <https://www.cloudflare.com/products/tunnel/>

THANK YOU

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